

ITA0606 - Machine Learning Assignment 2

Q1. Functional Blocks of a Machine Learning System

Architecture: Data Collection → Preprocessing → Feature Extraction → Model Training → Evaluation → Deployment.

Data Collection: Gather data from sensors, databases or user logs.

Preprocessing: Handle missing values, remove noise, normalize data and encode categorical values.

Feature Extraction: Select informative attributes (e.g., edges in images, TF-IDF in text).

Model Training: Learn patterns using algorithms such as Decision Trees, SVM or Neural Networks.

Evaluation: Measure Accuracy, Precision, Recall and F1-score on test data.

Supervised Example: Email spam detection uses labelled emails (Spam/Not Spam).

Unsupervised Example: Customer segmentation using K-Means uses unlabelled data to discover groups.

Q3. Backpropagation in MLP

Forward pass computes outputs layer by layer using weighted sums and activation functions. Error = Target – Predicted. Backpropagation computes gradients using the chain rule and updates weights: $W(\text{new}) = W(\text{old}) - \eta \times \partial \text{Loss} / \partial W$. Example: Target=1, Output=0.8, Error=0.2. Gradients are propagated backward from output to hidden layer. A proper learning rate ensures stable convergence. Sigmoid, ReLU and Tanh influence convergence speed and accuracy.

Q6. Decision Tree Learning

Decision Tree consists of Root Node, Internal Nodes, Branches and Leaf Nodes. Feature selection uses **Entropy** and **Information Gain**. The attribute with highest information gain becomes the root. Example: Outlook → Sunny/Rainy. Sunny branches split further using Humidity. Leaves contain final class labels. Pruning removes unnecessary branches after training to reduce overfitting and improve generalization.

Q10. Classification Evaluation Metrics

Sample confusion matrix: TP=45, TN=40, FP=5, FN=10.

Accuracy = $(TP + TN) / (\text{Total}) = 85\%$.

Precision = $TP / (TP + FP) = 90\%$.

Recall = $TP / (TP + FN) = 81.8\%$.

F1-score = $2PR / (P + R) = 85.7\%$.

ROC curve plots True Positive Rate against False Positive Rate. High AUC indicates better classifier performance. Medical diagnosis emphasizes Recall, while fraud detection often prioritizes Precision.

Q11. Artificial Neural Network

ANN contains Input Layer, Hidden Layer(s) and Output Layer. Inputs are multiplied by weights, biases are added, and activation functions generate outputs. Forward propagation moves information from input to output. During training, backpropagation updates weights to minimize error. Overfitting is reduced using Dropout, L2 Regularization and Early Stopping.